FINITE SEMI-SEGAL COHERENCE FOR THE AMBIENT CATEGORY

Status. Mathematical proposal, preceding implementation. The target is the existing internal pentagon type, `HcompPentagon.Pentagon`, for `hcomp` and `hcomp-assoc`. No such proof is claimed here. The checked development, including its explicit `hpent` parameter and all homology constructions, is unchanged. An independent internal refactor has subsequently been checked in `unit/sag/hcat-segal`; see `hcat-segal-refactor.pdf`. It derives the pentagon for new operations from finite internal semi-Segal data and reuses the homology proofs. It does not establish the ambient construction proposed here.

1. The two coherence problems

There are two distinct categorical structures. The ambient one has objects Cat and arrows $\mathsf{Map}(A, B)$, namely functors. Inside $C : \mathsf{Cat}$, objects are $\mathsf{Map}(1, C)$ and morphisms are fibres of

$$\mathsf{Map}(I, C) \longrightarrow \mathsf{Map}(1, C) \times \mathsf{Map}(1, C), \qquad m \longmapsto (m \circ I_0, m \circ I_1).$$

An internal morphism therefore includes its endpoint paths. Equality of its underlying map $I \to C$ alone does not establish equality in Hom_C.

The current ambient parameters give composition, associativity and unitors in `Wild`, the pentagon and triangle in `Coh2`, and a specified K_5 equation in `Coh3`. These are ordinary identity types; functor composition has no special conversion rules. The internal composition and associator are constructed using the geometric simplices $\Delta^n : \mathsf{Cat}$. Although

$$\mathsf{Map}(\Delta^n, C) \simeq \mathsf{Chains}_n(C)$$

is already proved for every n, the compatibility of restrictions and functor actions still uses ambient coherence. This family of equivalences alone is not a constructed semisimplicial object with all matching data.

The present obstruction is concrete. `HomTripleNormalize` expresses coherence of the existing `Hom-map-of-comp` as two endpoint equations, `LeftCoherence` and `RightCoherence`. `PentagonReduction.assoc-on-spine` already identifies the chosen associator on a 3-simplex with its defining pasting, and `PentagonReduction.extend` reduces the desired pentagon to 4-simplex spines. These comparisons should be reused.

2. A finite ambient presentation

Fix the bound 5 externally. Define a telescope of families

$$N_0 : \mathcal{U}_{l+1}, \qquad N_1(A, B) : \mathcal{U}_l, \qquad N_r : M_r(N_{<r}) \to \mathcal{U}_l \quad (2 \le r \le 5),$$

with $N_0 = \mathsf{Cat}$ and $N_1(A, B) = \mathsf{Map}(A, B)$. For each of the four displayed values of r, M_r is an explicitly expanded dependent sum of proper faces of an r-simplex. A boundary has one component x_S for each nonempty proper subset $S \subsetneq \{0, \dots, r\}$, in increasing cardinality, with

$$x_S : N_{|S|-1}\big((x_T)_{\varnothing \neq T \subsetneq S}\big).$$

This is notation for four finite expressions, not a claimed internal recursive definition of all semisimplicial types. The total boundary types may be large because their vertices lie in Cat; the filler families remain locally small.

Faces are projections of these tuples. Their shared subfaces are shared variables. This use of the ordinary computation rules for dependent pairs introduces no judgmental equation for composition

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in Cat. In particular, a nerve whose simplices are just chains, with inner faces defined by composition, would not have this property: its face identities are precisely part of the problem.

For $2 \le r \le 5$, let $\Lambda^r_1(N)$ be the same telescope with the top face and the face opposite vertex 1 omitted. For a horn h, require

$$\mathsf{isContr}\left(\sum_{u:N_{r-1}(\partial_1 h)} N_r(h,u)\right). \tag{1}$$

Here $\partial_1 h$ is the already specified boundary of the missing face; (h,u) means its insertion in the full boundary in the prescribed order. The requirement concerns the *missing face together with the top filler*. It does not assert that every full boundary has a filler, nor that $N_r(b)$ is contractible for every boundary b.

Proposal 1. Present the ambient associative coherence by a 5-restricted semi-Segal type as above, retaining the existing structural axioms for Cat. For the initial experiment, extend a normalized 4-restricted presentation of the current `Coh2` structure, so that all existing choices are fixed up to specified dependent comparisons. Supply the new coherence by the degree-5 family and (1), rather than by a separate associahedral equation.

This changes the presentation of the intended external coherence schema; it does not derive an extension from arbitrary `Coh2` data. In particular, existence of the degree-5 extension is an input, not a consequence of contractibility in degrees at most 4. Its equivalence with the precise existing K_5 parameter is a theorem to establish, not an assumption of this note.

3. Normalization, units, and the dimension bound

Normalization makes compatibility with the old operations a visible part of the task. For example take

$$N_2(f_{01}, f_{12}, f_{02}) := \mathsf{Id}(f_{12} \circ f_{01}, f_{02}).$$

Its 2-horn fillers are a singleton, with centre $(f_{12} \circ f_{01}, \mathsf{refl})$. For a tetrahedron write $q_{ijk} : f_{jk} \circ f_{ij} = f_{ik}$. Use the current reassociator $a(h,g,f) : (h \circ g) \circ f = h \circ (g \circ f)$ and set N_3 to the identity type between the two paths

$$L = \mathsf{ap}_{(-\circ f_{01})}(q_{123}) \cdot q_{013},$$
$$R = a(f_{23}, f_{12}, f_{01}) \cdot \mathsf{ap}_{(f_{23}\circ -)}(q_{012}) \cdot q_{023}.$$

Both run from $(f_{23} \circ f_{12}) \circ f_{01}$ to f_{03}. For a Λ^3_1 horn the unknown q_{023} can be solved for; cancellation of paths proves contractibility of its paired filler. Thus specifying a tetrahedron is specifying an associativity comparison with all its boundary witnesses, not replacing that comparison by reflexivity.

The degree-4 normalization similarly encodes the supplied ambient pentagon, including arbitrary triangle and tetrahedron boundaries. The finite comparison with wild 2-semicategories supplies the method for constructing this presentation. A comparison with the actual `pentagonator`, including its orientation and path bracketing, must accompany it. Keeping only agreement of binary composition is insufficient.

Semisimplicial data alone have no units. For this experiment retain `idMap`, the two unitors and `triangle2`. They can also be represented by degeneracy data in dimensions 1, 2 and 3 on the normalized presentation: identity edges, unit triangles, and degenerate tetrahedra. The latter give unit coherence. We need neither a claim that identities follow from the nonunital horn condition nor an unbounded degeneracy structure. The existing `cat-univalence` and internal `rezk` parameters remain in place. Completeness should be compared with them only after unit compatibility is proved.

Ambient degree	Extracted comparison	Existing use
2	binary composition	`comp`
3	associator	endpoint correction in `Hom-map`
4	pentagon	`Hom-map-of-comp`
5	coherence between pentagon pastings	triple functor action

The last row is the proposed new extraction lemma. Its dimension is motivated by the actual endpoint calculations. For $K : A \to B$, $G : B \to C$, $F : C \to D$ and $m : I \to A$, the underlying arrow comparison involves four maps (F, G, K, m), but evaluating an endpoint involves five:

$$1 \xrightarrow{I_\epsilon} I \xrightarrow{m} A \xrightarrow{K} B \xrightarrow{G} C \xrightarrow{F} D, \qquad \epsilon = 0, 1.$$

This explains why an ambient degree-4 presentation should not simply be declared sufficient for the current proof. Degree 5 is a proposed sufficient bound for this route, not a proved minimal bound. Internal degree 4 and ambient degree 5 refer to different simplices.

4. The route to the specified internal pentagon

The first test of the presentation should be a theorem producing `HomTriple.Triple.Coherence` from the ambient degree-5 extension. For the canonical morphism $(m, \mathsf{refl}, \mathsf{refl})$, build the two normalized endpoint pastings from restrictions of a common ambient 5-simplex. Their common boundary and their identifications with `LeftCoherence` and `RightCoherence` must be written down. Then apply `from-endpoints`. Identity elimination extends the result to arbitrary endpoint witnesses.

One can either extract the particular K_5 equality first and finish the existing normalization argument, or compare the endpoint pastings directly as dependent restrictions of horn fillers. The latter avoids making K_5 a public axiom. In both cases a common contractible filler space is useful only *after* the specified pastings have been lifted to it with their boundary paths. Contractibility cannot supply these identifications for free.

The remaining proof obligations are the following.

(1) Lift triple functor-action coherence to the triangle boundary type. The vertex comparisons use ambient degree 4; the edge comparisons use the preceding endpoint theorem. Retain all boundary paths in this lift.
(2) Prove that the current composition-preservation witnesses for FG agree with the composite of those for G and F, with the `Hom-map-of-comp` corrections. Compare the full 2-simplex fillers, including their spine equalities, before projecting to composition witnesses.
(3) Prove preservation of the current `hcomp-assoc` by a functor. Induct along `segal3`; use `assoc-on-spine` to expose exactly `OnSpine.result`. The preceding witness comparison supplies the needed compatibility of its five factors.
(4) For $T : \mathsf{Map}(\Delta^4, C)$, apply this preservation theorem to the five associators in the internal pentagon. After the composition-preservation corrections cancel, the two paths come from parallel paths in a hom-type of Δ^4. Those hom-types are propositions, so their parallel paths agree. Apply `PentagonReduction.extend` to obtain the pentagon on every four-arrow chain.

These are proof tasks, not new parameter fields. In particular, neither the endpoint theorem nor preservation of associators should be assumed in the definition of the proposed ambient structure.

An alternative packaging is to construct a 4-restricted internal semi-Segal type S_C whose total simplex types agree with $\mathsf{Map}(\Delta^r, C)$, for $0 \le r \le 4$, compatibly with restrictions and the existing spine maps. The already proved Segal equivalences would then supply its filler contractibility. The word "compatibly" carries the work: independent levelwise equivalences do not suffice to construct S_C.

Proposition 1 (A sufficient internal interface)**.** *Suppose such an S_C has its edge family identified with the existing Hom_C, and its extracted binary composition and associator identified with the*

existing ones as a dependent pair of operations and laws. Then the pentagon extracted from S_C transports to `HcompPentagon.Pentagon` *for the existing operations.*

Proof. A 4-restricted semi-Segal structure supplies a pentagon for its extracted composition and associator. Transport this term along the stated dependent identification. Equivalently, eliminate the identification first; the target is then exactly the extracted pentagon type. □

The requirement concerning the associator cannot be dropped: equality of operations does not identify two chosen paths expressing their associativity in an untruncated hom-type. This proposition isolates a precise sufficient interface; constructing it from the ambient degree-5 data is the proposed program above.

5. Comparison with the literature

Sattler–Wärn explicitly mention a Reedy fibrant complete semi-Segal presentation of the ambient wild category before adopting judgmental associativity and unit laws in Axiom 1(5)–(6) of their HoTT/UF abstract [1]. The present proposal follows the schematic alternative, retaining their structural axioms and making a finite bound explicit. Wärn's local *Higher-categorical headcanon*, Axiom 1 and Remark 2, likewise treat infinite coherence as an external schema; Remark 32 emphasizes that constructed internal laws must retain their witnesses to extend to higher laws. This is precisely why preservation of the chosen associator matters here.

Capriotti–Kraus prove the equivalence between 4-restricted semi-Segal types and wild 2-semicategories, before imposing truncation [2, Theorem 3.6]. Their later complete semi-Segal n-types include an $(n-1)$-truncation condition on hom-types; we do not use that condition for $\mathsf{Map}(A, B)$. Their finite dependent-family encoding and low-dimensional unit comparison apply to the proposed normalization. The degree-5 extraction and the comparison with this repository's internal associator go beyond the stated degree-4 result. Completeness is useful for univalence and recovering identities; it does not remove the need for those comparisons.

Kraus–Sattler's work on space-valued diagrams provides finite diagram constructions and a way of incorporating degeneracies [3]. That is relevant if we later need a systematically constructed finite simplicial presentation, beyond the unit fragment used here. Annenkov–Capriotti–Kraus–Sattler use two-level type theory to express externally indexed constructions internally [4]. Here the bound is fixed before writing the telescope, so no outer equality, UIP, or two-level extension is being used. Sattler's marked semisimplicial *sets* give a constructive semantic model of higher categories [5]; that is different from the dependent-family presentation of the untruncated mapping types here.

6. Integration boundary and next experiment

Start in a separate checking root with the finite matching telescopes, the degree-2/3 normalization, and the degree-4 comparison to the existing `Coh2` data. Next add degree 5 and attempt the endpoint theorem above. This tests the proposed bound at the current obstruction before undertaking an internal nerve construction.

Do not change `All.seg`, `Composition`, `Associativity`, or `Homology.WithPentagon` during this experiment. Once a term of the exact existing pentagon type has been checked, instantiate `WithPentagon` with it. That preserves the existing groups, structural connecting homomorphism, positive-degree exact sequence, and sphere calculations. Degree-zero extension and categorical naturality remain separate open homology tasks.

No new axioms have been installed by this note. The current `Theory` parameters remain `univ`, `images`, and `surjections`; all current Cat, copower, zero-object, suspension and stability parameters, including `hpent`, remain unchanged. Adopting the proposal will require an explicit finite ambient extension parameter, together with proved comparisons. It must not be reported as a consequence of the current level-2 parameters alone, nor as a completed derivation of `hpent`.

References

[1] C. Sattler and D. Wärn, *Axioms for higher category theory*, HoTT/UF 2026 abstract. https://hott-uf.github.io/2026/abstracts/HoTTUF_2026_paper_20.pdf.

[2] P. Capriotti and N. Kraus, *Univalent Higher Categories via Complete Semi-Segal Types*, full version, especially Sections 2–4. https://arxiv.org/abs/1707.03693.

[3] N. Kraus and C. Sattler, *Space-Valued Diagrams, Type-Theoretically*, extended abstract. https://arxiv.org/abs/1704.04543.

[4] D. Annenkov, P. Capriotti, N. Kraus and C. Sattler, *Two-Level Type Theory and Applications*. https://arxiv.org/abs/1705.03307.

[5] C. Sattler, *Constructive homotopy theory of marked semisimplicial sets*. https://arxiv.org/abs/1809.11168.